

● RAG Learning

4.2 Congruence, Similarity & Transformations

AQA GCSE Mathematics 8300 — Geometry and Measures — spec refs G5, G7, G8, G19 — Papers 1 (non-calculator), 2 and 3 (calculator)

This topic is about how shapes can be moved, flipped, turned and resized, and how to tell when two shapes are identical (**congruent**) or the same shape but a different size (**similar**). Higher tier adds negative scale factors, combined transformations, invariance, and the link between length, area and volume in similar figures.

4.2.1 Congruent Triangles (G5)

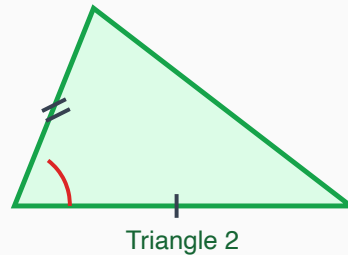
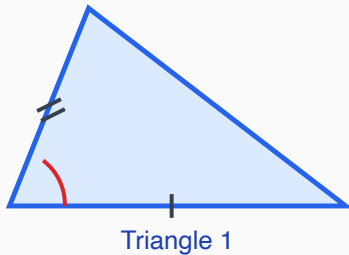
Two shapes are **congruent** if they are exactly the same shape *and* size. One can be mapped onto the other by a translation, rotation or reflection (or a combination of these). All corresponding sides and all corresponding angles are equal.

You do not need to check all six measurements to prove two triangles congruent. Any one of these four conditions is enough:

Condition	What must be equal	Notes
SSS	All three sides	Side, Side, Side
SAS	Two sides and the angle <i>between</i> them	The angle must be the included angle
ASA (or AAS)	Two angles and a corresponding side	If two angles match, the third does too, so any corresponding side works

RHS	Right angle, Hypotenuse and one other Side	Only for right-angled triangles
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SAS: two marked sides and the included angle match, so the triangles are congruent



Exam Tip: **AAA is not a congruence condition.** Three equal angles only prove the triangles are *similar* — one could be an enlargement of the other. Similarly, **SSA** (two sides and a non-included angle) is not enough, except in the special right-angled case **RHS**.

Writing a congruence proof

A congruence proof is a short chain of statements, each with a **reason**, ending with the condition used.

Worked Example: ABCD is a parallelogram. Prove that triangles ABD and CDB are congruent.

$AB = CD$ (opposite sides of a parallelogram are equal)

$AD = CB$ (opposite sides of a parallelogram are equal)

BD is common to both triangles.

So triangles ABD and CDB are **congruent (SSS)**.

Worked Example: ABC is isosceles with $AB = AC$, and M is the midpoint of BC. Prove that the line AM is perpendicular to BC.

$AB = AC$ (given), $BM = CM$ (M is the midpoint), AM is common, so triangles ABM

and $\triangle ACM$ are congruent (SSS).

Hence angle $\angle AMB = \angle AMC$. These two angles lie on a straight line, so each is $180^\circ \div 2 = 90^\circ$. So $AM \perp BC$.

4.2.2 The Four Transformations (G7, G8)

A **transformation** maps an *object* onto an *image*. In the exam you must be able to carry out each one on coordinate axes and to **describe it fully**. "Describe fully" questions usually carry 2–3 marks, one for the name and one for each piece of detail.

Transformation	To describe it fully, give...	Image congruent?
Translation	the column vector	Yes
Reflection	the equation of the mirror line	Yes
Rotation	the centre, the angle, the direction	Yes
Enlargement	the scale factor and the centre	No — similar (unless scale factor is 1 or -1)

Translation

A **translation** slides every point the same distance in the same direction. It is written as a **column vector** $\begin{pmatrix} x \\ y \end{pmatrix}$: the top number is the movement right (negative = left), the bottom number is the movement up (negative = down).

Worked Example: Translate the triangle with vertices $(1, 1)$, $(3, 1)$, $(3, 2)$ by $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$.

Subtract 4 from each x and add 3 to each y : $(-3, 4)$, $(-1, 4)$, $(-1, 5)$.

Reflection

Every point is the same perpendicular distance from the mirror line as its image, on the opposite side. Common mirror lines:

Reflection in the x -axis ($y = 0$): $(x, y) \rightarrow (x, -y)$

Reflection in the y -axis ($x = 0$): $(x, y) \rightarrow (-x, y)$

Reflection in $y = x$: $(x, y) \rightarrow (y, x)$

Reflection in $y = -x$: $(x, y) \rightarrow (-y, -x)$

Reflection in $x = a$: the x -coordinate becomes $2a - x$ e.g. $(5, 3)$ in $x = 2$ gives $(-1, 3)$

Rotation

A rotation turns a shape about a fixed **centre of rotation**. You need tracing paper in the exam — ask for it. Rotations of 90° and 180° about the origin follow simple rules:

90° clockwise about O : $(x, y) \rightarrow (y, -x)$

90° anticlockwise about O : $(x, y) \rightarrow (-y, x)$

180° about O : $(x, y) \rightarrow (-x, -y)$ (direction does not matter for 180°)

Worked Example: Rotate $(5, 1)$ by 90° anticlockwise about $(2, 1)$.

Position relative to the centre: $(5 - 2, 1 - 1) = (3, 0)$.

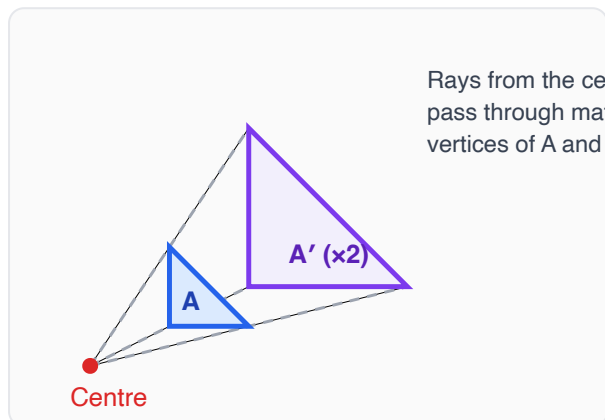
Rotate 90° anticlockwise: $(3, 0) \rightarrow (0, 3)$.

Add the centre back: $(2 + 0, 1 + 3) = (2, 4)$.

Exam Tip: A rotation description needs **three** things: centre, angle and direction. Missing the direction on a 90° rotation loses a mark. Never describe a single transformation as a combination (e.g. "reflect then translate") — that scores zero.

4.2.3 Enlargement and Scale Factors (G7)

An **enlargement** changes the size of a shape. Every length is multiplied by the **scale factor** k , and every point moves along a line from the **centre of enlargement** so that its distance from the centre is multiplied by k . Angles do not change, so the image is **similar** to the object.



Scale factor k	Effect on the image
$k > 1$	Larger, further from the centre
$0 < k < 1$ (fractional)	Smaller, closer to the centre — still called an enlargement
$k < 0$ Higher	On the opposite side of the centre and upside down

Image of point P under enlargement, scale factor k , centre C :

$$P' = C + k(P - C)$$

$$\text{Scale factor} = \frac{\text{image length}}{\text{object length}}$$

Worked Example: Enlarge $(6, 4)$ and $(10, 8)$ by scale factor $\frac{1}{2}$, centre $(2, 2)$.

$(6, 4)$: vector from centre = $(4, 2)$; halve it: $(2, 1)$; image = $(2 + 2, 2 + 1) = (4, 3)$.

$(10, 8)$: vector from centre = $(8, 6)$; halve it: $(4, 3)$; image = $(6, 5)$.

Finding the centre of enlargement

Draw straight lines through each pair of corresponding vertices (object vertex to image vertex) and extend them. They all cross at the centre. Algebraically, rearrange $P' = C + k(P - C)$ to get

$$C = \frac{P' - kP}{1 - k}.$$

Worked Example: Triangle A has vertices $(2, 1)$, $(3, 1)$, $(2, 3)$; triangle B has vertices $(3, 2)$, $(5, 2)$, $(3, 6)$. Describe the transformation.

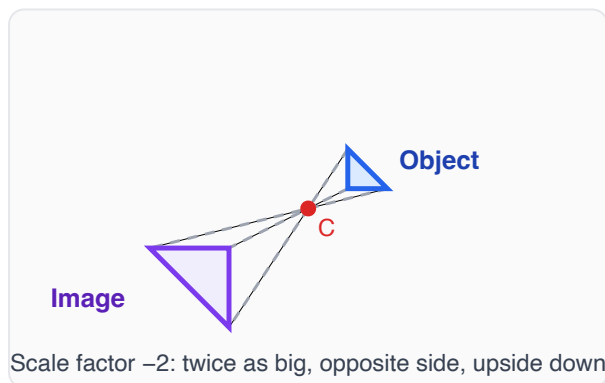
Side $(2, 1)-(3, 1)$ has length 1; the matching side of B has length 2, so $k = 2$.

$$\text{Centre: } C = \frac{(3, 2) - 2(2, 1)}{1 - 2} = \frac{(-1, 0)}{-1} = (1, 0).$$

Enlargement, scale factor 2, centre $(1, 0)$.

Negative scale factors Higher

With a negative scale factor, the image is on the *opposite* side of the centre. Measure from the centre to each object vertex, multiply by $|k|$, and go that distance in the *opposite* direction. The image appears rotated by 180° .



Worked Example: Enlarge the triangle $(2, 1)$, $(4, 1)$, $(2, 4)$ by scale factor $-\frac{1}{2}$, centre O .

Multiply each coordinate by $-\frac{1}{2}$: $(-1, -0.5)$, $(-2, -0.5)$, $(-1, -2)$.

Exam Tip: An enlargement with scale factor -1 is the same as a **rotation of 180°** about the centre of enlargement. If asked to describe a single transformation, either answer earns the marks provided the centre is correct.

4.2.4 Combined Transformations and Invariance Higher

When one transformation is followed by another, the result can often be described as a **single** transformation. Work through the coordinates of each vertex step by step, then compare the final image with the original.

Worked Example: Triangle T $(1, 2)$, $(3, 2)$, $(3, 3)$ is reflected in $y = x$, then the image is reflected in the x -axis. Describe the single transformation.

Reflect in $y = x$: $(x, y) \rightarrow (y, x)$ gives $(2, 1)$, $(2, 3)$, $(3, 3)$.

Reflect in x -axis: $(x, y) \rightarrow (x, -y)$ gives $(2, -1)$, $(2, -3)$, $(3, -3)$.

Overall $(x, y) \rightarrow (y, -x)$: **rotation 90° clockwise about the origin.**

An **invariant point** is a point that maps onto itself. An **invariant line** maps onto itself (its points may move along it).

- Reflection: every point on the mirror line is invariant.
- Rotation: only the centre of rotation is invariant.
- Enlargement ($k \neq 1$): only the centre is invariant.
- Translation (non-zero vector): no invariant points.

Useful results about combinations:

- Two reflections in **parallel** lines give a **translation** (twice the distance between the lines).
- Two reflections in lines that **cross** give a **rotation** about the crossing point, through twice the angle between the lines.

- Two translations combine by adding the column vectors: $\begin{pmatrix} 2 \\ 5 \end{pmatrix} + \begin{pmatrix} -6 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$.

- Order can matter: "reflect then rotate" may not give the same image as "rotate then reflect".

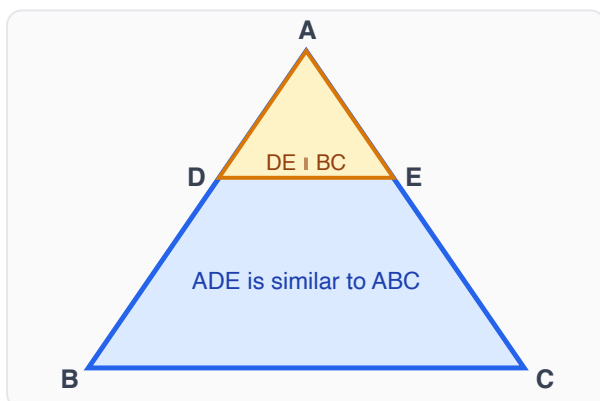
4.2.5 Similar Shapes (G7, G19)

Two shapes are **mathematically similar** if one is an enlargement of the other.

Corresponding **angles are equal** and corresponding **sides are in the same ratio** (the length scale factor k).

Two triangles are similar if any one of these is true:

- all three pairs of angles are equal (two pairs is enough, as the third must then match);
- all three pairs of sides are in the same ratio;
- two pairs of sides are in the same ratio and the included angles are equal.



When a line inside a triangle is parallel to one side (DE parallel to BC above), it cuts off a smaller similar triangle: angle A is shared, and the other angles are equal as corresponding angles.

Worked Example: In the diagram, $AD = 4$ cm, $DB = 6$ cm and $DE = 5$ cm. Find BC .

$AB = 4 + 6 = 10$ cm, so the scale factor from ADE to ABC is $10 \div 4 = 2.5$.

$BC = 5 \times 2.5 = 12.5$ cm.

Exam Tip: Use the **whole** side of the large triangle ($AB = AD + DB$), not just the extra piece DB . Using $6 \div 4$ is the most common mistake. Sketch the two triangles separately,

matching equal angles, before finding the scale factor.

Worked Example: Lines AE and BD cross at C , with AB parallel to DE . $AB = 9$ cm, $DE = 6$ cm, $AC = 7.5$ cm, $CD = 4$ cm. Find CE and BC .

Angle $ACB = \text{angle } ECD$ (vertically opposite); angle $BAC = \text{angle } DEC$ (alternate). So ABC and EDC are similar with $A \leftrightarrow E, B \leftrightarrow D$.

Scale factor (small to large) $= 9 \div 6 = 1.5$.

$CE = 7.5 \div 1.5 = 5$ cm; $BC = 4 \times 1.5 = 6$ cm.

4.2.6 Lengths, Areas and Volumes of Similar Figures Higher

If the length scale factor between two similar shapes is k , areas are multiplied by k^2 and volumes by k^3 . This applies to *any* area (surface area, cross-section, area of a label) and *any* volume (capacity, and mass if the material is the same).

Length scale factor $= k$

Area scale factor $= k^2$

Volume scale factor $= k^3$

Going backwards: $k = \sqrt{\text{area scale factor}} = \sqrt[3]{\text{volume scale factor}}$

Length ratio	Area ratio	Volume ratio
1 : 2	1 : 4	1 : 8
1 : 3	1 : 9	1 : 27
2 : 3	4 : 9	8 : 27
3 : 5	9 : 25	27 : 125

Worked Example: Two similar bottles are 15 cm and 25 cm tall. The smaller holds 270 ml. How much does the larger hold?

$$k = \frac{25}{15} = \frac{5}{3}, \text{ volume scale factor} = \left(\frac{5}{3}\right)^3 = \frac{125}{27}.$$

$$\text{Capacity} = 270 \times \frac{125}{27} = \mathbf{1250 \text{ ml.}}$$

Worked Example: Two similar pyramids have volumes 54 cm^3 and 250 cm^3 . The smaller has surface area 36 cm^2 . Find the surface area of the larger.

$$\text{Volume scale factor} = \frac{250}{54} = \frac{125}{27}, \text{ so } k = \sqrt[3]{\frac{125}{27}} = \frac{5}{3}.$$

$$\text{Area scale factor} = \left(\frac{5}{3}\right)^2 = \frac{25}{9}.$$

$$\text{Surface area} = 36 \times \frac{25}{9} = \mathbf{100 \text{ cm}^2}.$$

Exam Tip: Always go through the **length** scale factor. If you are given areas and need a volume, square-root to get k , then cube it. Never multiply an area by a length scale factor directly.

Summary

Idea	Key fact
Congruent triangles	SSS, SAS, ASA/AAS, RHS (not AAA)
Translation	Column vector
Reflection	Equation of mirror line
Rotation	Centre, angle, direction

Enlargement	Scale factor, centre; fractional k makes smaller
Negative scale factor Higher	Opposite side of centre, inverted
Invariant points Higher	Points that map to themselves
Similar figures Higher	Lengths $\times k$, areas $\times k^2$, volumes $\times k^3$